

Figure 1 | What does causal-role recurrence preserve?

A Checkpoint-local reconstruction and prospective intervention



B Dominant ranks are local to each checkpoint

Mamba-130M
P3

Mamba-370M
P3

Mamba-1.4B
P5

Equal rank labels do not imply semantic identity.

C Across the tested checkpoints

Causal-role recurrence does not guarantee geometric invariance
or downstream functional invariance.